How Much of the Learned Testing Policy Should a Model Keep?

Adapting deep clinical time-series models to observation shift from a small run-in

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Advisor: Sarah Martinson · Topic: Deep Time-Series Models under Observation Shift

1 Project idea and research question

Deep models for clinical time series learn from *what* is measured and also from *when* and *whether* it is measured: a lactate order signals that a doctor suspects sepsis. This testing pattern is predictive [1], but it reflects the hospital's observation policy, and it changes when a model trained at hospital A is deployed at hospital B (*observation shift*). The change comes in two kinds. Hospital B may test everyone more, so every order means a bit less; or individual tests may change meaning, for example troponin is ordered for 3% of patients at Beth Israel but 32% at Emory (PhysioNet 2019 [2]), while a lactate order may keep its meaning. A model adapted to B must fix the first kind globally and the second per test.

Before a model goes live, hospitals often run it silently for 1–3 months [3]. That run-in yields the only labelled data from B, and very little of it: a typical ICU admits 600–1,300 patients a year [4], so a 1–3 month run-in contains about 2–6 sepsis cases. We therefore propose a **ladder of adaptations** that changes only the testing part of a model whose physiology and testing evidence are kept separate: (1) one global weight w on the testing evidence (w=0 ignore, w=1 trust as at A); (2) one weight per group of tests; (3) re-learning the testing part with physiology frozen; (4) fine-tuning the whole model. Higher rungs repair more kinds of shift but need more outcome cases.

Research question. *For a run-in of a given size at a new hospital, which rung of adaptation of the testing part predicts best, and how does the answer depend on the kind of observation shift?* Concretely: (i) does adapting only the testing part beat the fixed rules of prior work (always keep or always remove the pattern) and full fine-tuning at realistic run-in sizes; (ii) at how many outcome events does each higher rung start to pay off; (iii) does a model with a separate testing part adapt better from few cases than a standard architecture (GRU-D) fine-tuned at its last layer or in full.

2 Literature overview

Deep models use the observation pattern but do not separate it. Architectures for irregular series consume masks and time gaps (GRU-D [5]), continuous-time paths (Neural CDE [6], Latent ODE [7]) or time-value sets and attention (mTAN [8], Raindrop [9]), and fuse the pattern with physiology in one representation. Models with a separate observation component exist, a Hawkes process for clinicians' measurement events [10] and DeepJoint's observation heads [11], but none adapts that component alone at a new site.

Observation shift between hospitals. Multi-site deep sepsis and ICU benchmarks lose 0.08–0.09 AUROC across sites [12, 13] and include missingness indicators without attributing the loss to them. Studies that do isolate the pattern choose all or nothing: Futoma et al. [14] fitted with measurement indicators and dropped them at deployment, which transferred best from Beth Israel to a Memphis hospital but not in reverse; Rockenschaub et al. [15] found that ignoring informative missingness costs at the source but is more robust elsewhere, especially when missingness is highly informative. Lasko et al. [16] argue conceptually for a site-specific process model next to a stable disease model; Zhou et al. [17] adapt to missingness shift without labels in linear models. None tunes *how much* of the learned policy to keep from labelled target data.

Adapting with few labels. Clinical model updating is a ladder from recalibration to full refitting, and parsimonious updates win at small samples [18, 19]; a full refit is already risky well below the 100 events needed just to validate a model [20]. In deep learning, which part to fine-tune depends on the kind of shift [21], and retraining only the last layer can suffice [22], but clinical fine-tuning studies use hundreds to thousands of target patients [12, 13]. **Gap:** no work adapts only the observation part of a clinical sequence model, and none compares such adaptations against run-in size.

Preliminary results (linear models, open PhysioNet data). On Beth Israel → Emory, keeping the

pattern was best, because there was barely any observation shift between the two (pattern-model transfer gap 0.03 AUROC, 95% CI −0.01 to 0.07); dropping it cost 0.05 AUROC. Across 26 unit pairs the pattern's meaning shifted clearly in 7, mostly for mortality; there, a single weight tuned on a 3–24 month run-in gained up to 0.02–0.03 AUROC and stayed within 0.001 elsewhere (after removing impossible lab values). A one-parameter update beat a full refit up to about 20 cases; the refit won from about 50.

3 Plan of experiments

Model. A sequence model whose outcome logit is the sum of a physiology part (GRU-D encoder over values) and a testing part (encoder over masks, counts and time gaps); the rungs act on the testing part only. Baselines: keep (w=1), remove (w=0, Futoma), recalibration, and a standard GRU-D with last-layer or full fine-tuning. Lessons from our pilots are built in: tuned weights are shrunk towards 1, calibration is fitted separately from the weights, and impossible values are removed before recalibration.

E1 Simulation with known optimum. Our latent-state simulator, whose exact Bayes predictor is known, with two shift types: global (B tests less selectively) and test-specific (one test group becomes routine). Regret of each rung against the Bayes predictor at B, for run-ins of 5–200 outcome events. Expected: rung 1 wins under global shift and small run-ins, rungs 2–3 under test-specific shift once enough events exist.

E2 Real hospitals. The dataset is still open. Candidates: eICU-CRD (208 US hospitals, 200,000 ICU stays [23]; requires credentialed PhysioNet access), which gives many cross-hospital pairs; or the open PhysioNet 2019 (sepsis; two hospitals, four ICU units [2]) and 2012 (mortality; four ICU types) data of our pilots, which give 26 unit pairs. The design is fixed either way: ordered source → target pairs, run-ins of 1, 3, 6, 12 and 24 months drawn from the target site (20–200 draws), evaluation on all remaining target patients, primary metric AUROC (pre-specified), secondary calibration.

E3 Architecture. Whether the separated model adapts from fewer cases than GRU-D fine-tuned at the last layer or in full, on the pairs of E2.

Computational cost and feasibility. All models are small (GRU hidden size 32–64, at most 24–48 hourly steps per patient, a few thousand patients per site), so a CPU is enough; we have a 4-core laptop and can request ETH's Euler cluster if needed. Measured on the laptop: our simulation grid of 108 deep fits (GRU, GRU-D, Neural CDE, Transformer) took 3.6 hours; each real-data linear experiment took 3–11 minutes.

Component	Units	CPU-hours (estimate)
E1 simulation: source fits and all rungs	2 shifts × 6 run-in sizes × 20 draws	4
E2/E3 source models	8 sites × 2 models × 3 seeds	8
Rungs 1–2 (1–8 parameters each)	25 pairs × 5 sizes × 200 draws	<1
Rungs 3–4, last-layer tuning (∼15 s)	25 pairs × 5 sizes × 20 draws × 3	32
Total		≈45

The table assumes about 8 sites and 25 pairs, as in either candidate dataset; cost grows linearly with the number of pairs. About 45 CPU-hours run in about 12 hours on the laptop with four workers. The open PhysioNet data are already downloaded and processed, and the pilot pipeline exists. Timeline: weeks 1–2 the separated model and E1; weeks 3–4 E2 and E3; weeks 5–6 analysis and report.

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